Linear vs Nonlinear Models

Applied Linear Models I • Lecture 2 of 16

Where we are

- Lecture 1: a linear model is a **projection** of y onto $\mathcal{C}(X)$.
- Today: what makes a model "linear," and what to do when it is not.
- Key message: **"linear" means linear in the parameters**, not in the predictors.

Roadmap

Linearity in parameters $\rightarrow$ linearizable models $\rightarrow$ nonlinear least squares $\rightarrow$ how transformations move the error $\rightarrow$ log models and elasticity.

What does "linear" mean?

The mean response is a **linear combination of the parameters**:

$$\mathbb{E}[\mathsf{y}] = \mathsf{X}\boldsymbol{\beta} = \beta_0 \mathsf{x}_0 + \beta_1 \mathsf{x}_1 + \cdots$$

with regressors that do *not* depend on unknown parameters.

Operational test. A model is linear in β_j if $\partial \mathbb{E}[y]/\partial \beta_j$ is free of all parameters.

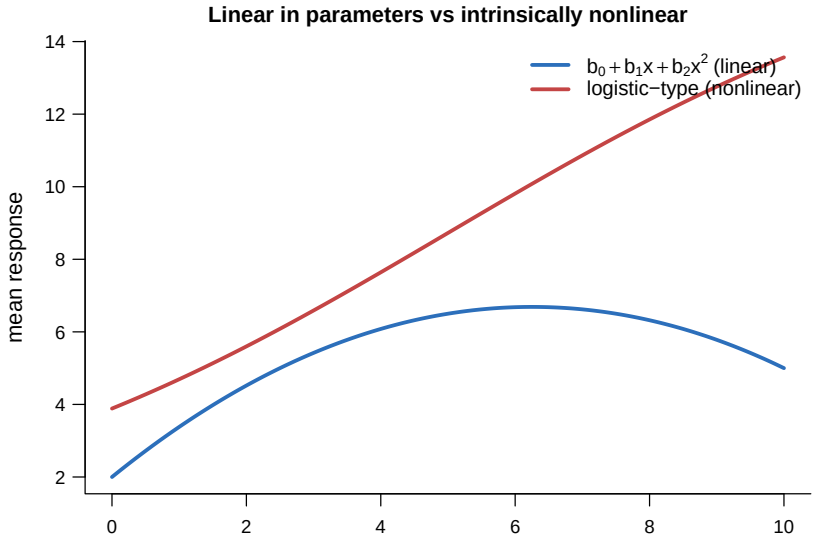
Linear

- $\beta_0 + \beta_1 x + \beta_2 x^2$
- $\beta_0 + \beta_1 \log x$
- $\beta_0 + \beta_1 \sin x$

Nonlinear

- $\beta_0\, e^{\beta_1 x}$
- $\dfrac{\beta_1 x}{\beta_2 + x}$
- $\beta_0 + \beta_1 \beta_2^x$

The predictor may be nonlinear; the model is not



Intrinsically linear (linearizable) models

Exponential growth looks nonlinear but transforms to linear:

$$y = a\,e^{bt} \quad \Longrightarrow \quad \log y = \underbrace{\log a}_{\beta_0} + \underbrace{b}_{\beta_1}\, t.$$

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But the transform decides the error model:

- Multiplicative error $y = ae^{bt}e^{\varepsilon} \Rightarrow \log y$ has additive normal error $\Rightarrow$ use OLS on logs.
- Additive error $y = ae^{bt} + \varepsilon \Rightarrow$ logs distort the error $\Rightarrow$ use nls on original scale.

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Worked result (multiplicative-error data, truth a=5, b=0.15)

Linearized OLS: $\hat{a}$=4.97, $\hat{b}$=0.151. nls: $\hat{a}$=5.30, $\hat{b}$=0.147.

Nonlinear least squares

When a model cannot be linearized, minimize the sum of squares directly:

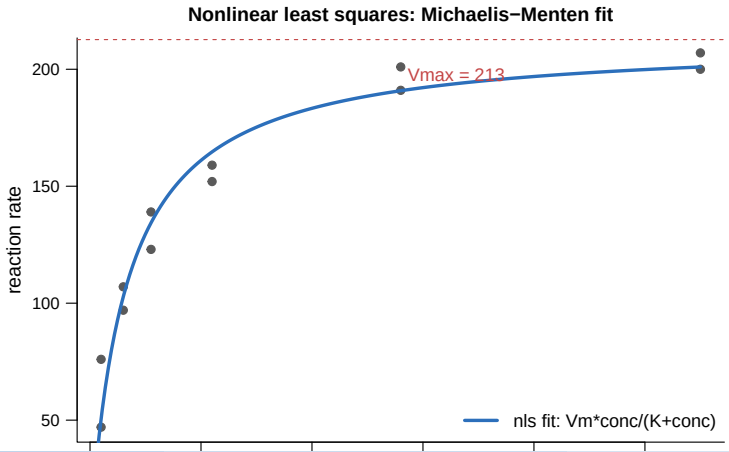
$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{n} \big(y_i - f(x_i; \boldsymbol{\theta})\big)^2,$$

solved *iteratively* (Gauss–Newton). In R: `nls()`; self-starting models (`SSmicmen`, `SSlogis`, ...) supply starting values.

- No closed form, only **asymptotic** inference.
- May have multiple local minima $\Rightarrow$ starting values matter.
- Residual structure and diagnostics still apply.

Example: Michaelis–Menten kinetics (Puromycin)

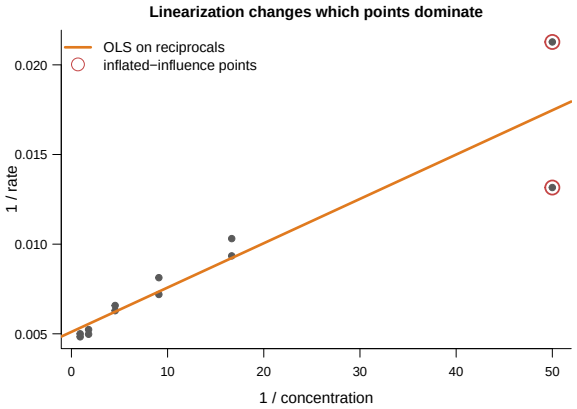
$$\text{rate} = \frac{V_{\max}\,\text{conc}}{K + \text{conc}}$$



Linearization can distort: Lineweaver–Burk

Michaelis–Menten rearranges to a straight line:

$$\frac{1}{\text{rate}} = \frac{1}{V_{\max}} + \frac{K}{V_{\max}} \cdot \frac{1}{\text{conc}}.$$



The reciprocal transform inflates the influence of small, noisy rates.

	$V_{\max}$	K
L–B (OLS)	196	0.048
nls (honest)	213	0.064

The estimates disagree: linearization preserves the mean, not the error.

Log models and elasticity

Model	Form	Slope means
Level–level	$y = \beta_0 + \beta_1 x$	$\Delta y = \beta_1$ per unit
Log–level	$\log y = \beta_0 + \beta_1 x$	$\approx 100\beta_1\%$ change in y per unit
Log–log	$\log y = \beta_0 + \beta_1 \log x$	elasticity: 1% in $x \to \beta_1\%$ in y

- Log–log slope is a *unit-free elasticity* – the econ workhorse (Cobb–Douglas, demand curves).
- Still **linear in the parameters** – the whole course applies.
- Worked example recovers elasticity $\hat{\beta}_1 = 0.70$.

Summary

- Linear = linear in the **parameters**; polynomials qualify, $\beta_0 e^{\beta_1 x}$ does not.
- Linearizable models can be transformed; intrinsically nonlinear ones need `nls`.
- **Transformations move the error** – transform only when the noise becomes well behaved on the new scale.
- Log–log slopes are elasticities and keep the model linear.

Next

Lecture 3: simple and multiple regression by OLS, and the Gauss–Markov theorem – why OLS is BLUE.